Abhishek Kumar

Software Engineer | Backend & Full-Stack | Robotics & IoT Systems
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Summary

Software Engineer with 1.5+ years of experience building production systems for robotics and enterprise teams. Built an edge data-logging and telematics system that captures CAN bus, motor controller, GNSS, radio and ROS topic traffic on autonomous vehicles, a microservices-based log analytics platform (RabbitMQ, Grafana), and a secure OTA firmware update platform (FastAPI, React, MQTT, PKI) for a 50+ robot fleet, all deployed on-premise with automated CI/CD. Independently built and deployed Learnova, a full-stack e-learning platform with web and mobile apps. Proficient in Python, REST API design, microservices, real-time messaging, Docker and Linux.

Experience

NewSpace Research & Technologies Bangalore, India
Junior Associate Software Engineer *Sep 2025 – Present*

- **Edge Data Logger:** Built an on-vehicle logging application that records all inter-subsystem communication, including **CAN bus** frames, **motor controller** telemetry, **GNSS** position data, **radio** link traffic and **ROS topics**, into a single time-synchronized log for debugging and post-mission analysis.
- Designed pluggable data-source adapters (CAN/SocketCAN, serial, UDP, ROS subscribers) with timestamping and buffered asynchronous writes so new subsystems can be logged without changing the core pipeline.
- Built the **telematics** sync layer that automatically uploads collected logs from vehicles to an on-premise server once connectivity is available, with no manual data transfer.
- **Logs & Telemetry Analytics Platform:** Developed a web application on a **microservices architecture (10+ services)** that ingests synced vehicle logs and lets engineers search, visualize and analyze them for debugging and post-mission review.
- Used **RabbitMQ** for asynchronous messaging between ingestion, parsing and storage services, and built interactive analysis dashboards in **Grafana** using the **Grafana Scenes** plugin framework.
- **Project Iris (OTA Platform):** Led architecture and development of a remote firmware update system for a fleet of **50+ robotic units** using **FastAPI** and **React**, removing the need for on-site technician visits.
- Implemented a real-time control layer over **MQTT** (QoS-based delivery) and **WebSockets** to push updates and stream live deployment logs from each robot.
- Secured OTA delivery with **PKI**-based firmware signing and encryption plus **mTLS**, and built **RBAC** (Admin / Technician / Viewer) with **JWT** authentication.
- Containerized the full stack (FastAPI, React, Mosquitto, PostgreSQL) with **Docker Compose**, cutting new-developer setup time by **80%**.
- **On-Premise Deployment & CI/CD:** Deployed the edge logger, analytics platform and Iris on in-office servers for data security, with end-to-end **CI/CD pipelines** that automatically build **Docker** images and deploy them on every merge.

RedAnt Technologies Bangalore, India
Junior Software Engineer *Jan 2025 – Sep 2025*

- Automated **5+** HR and admin workflows (attendance, leave, reimbursement, travel) using **Power Automate**, reducing processing time by **70%** and saving **15+ hours/week**.
- Engineered a dynamic Job Description generator that cut JD creation time from **2 hours to 5 minutes**.
- Built a centralized **SharePoint** intranet serving **200+ employees** as the single source of truth for policies and forms.

Projects

Learnova – E-Learning Platform (Web + Mobile) | *React, React Native (Expo), Node.js, Express, Prisma, PostgreSQL,*

- Built and deployed a full-stack course platform with a **React** web app (Vite, Tailwind CSS, Zustand) and a **React Native / Expo** mobile app (TanStack Query, NativeWind), both served by one **REST API** on **Node.js, Express, Prisma and PostgreSQL**.
- Implemented three roles (Student, Educator, Admin) with **JWT** authentication and **RBAC** middleware, plus an admin moderation workflow (Draft, Pending Review, Published, Rejected) and user blocking.
- Created a drag-and-drop course builder for sections and lessons supporting video, audio, PDF, rich-text articles (TipTap) and links, with direct browser uploads to **Cloudflare R2** via **S3 presigned URLs**.
- Added email **OTP** verification and password reset (Nodemailer), bcrypt hashing, enumeration-safe login, and per-lesson progress tracking with "continue learning"; evolved the schema through **15 Prisma migrations**.

Textsmith – Browser-Based Text & Developer Toolkit | *React.js, React Router, Bootstrap, Jest* | Live Demo

- Built a single-page app with **80+ text and developer tools** (regex find & replace, JSON/XML/HTML/SQL formatting, encoding, case and any-to-any number base conversion) that users can chain into a pipeline with a live preview.

- Runs entirely in the browser so no user data is uploaded, replacing many single-purpose websites with one fast, private workspace.
- Wrote all parsers and formatters from scratch with no external libraries, covered by **60+ automated Jest tests**, plus searchable docs with live examples, dark mode and saved sessions.

AI Chat Assistant | *TypeScript, React, Material UI, Node.js, Express, MongoDB, OpenAI API* | GitHub 2024

- Built a full-stack **GPT**-powered chat app in **TypeScript** (MERN stack) where each user's conversation history is stored in **MongoDB** and sent as context to the **OpenAI API**.
- Secured sessions with **JWT** in signed HTTP-only cookies and **express-validator** input checks; limited context to the last 10 messages to control token cost and added automatic retry on API rate limits (HTTP 429).

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, C++, SQL, Bash
Backend: FastAPI, Node.js, Express.js, REST API Design, Prisma ORM, Pydantic, AsyncIO, Microservices, RabbitMQ, OpenAI API
Frontend & Mobile: React.js, React Native (Expo), Vite, React Router, Redux Toolkit, Zustand, TanStack Query, Tailwind CSS, Material UI, Bootstrap, HTML5, CSS3
Robotics & IoT: ROS, CAN Bus (SocketCAN), GNSS/NMEA, Serial, UDP, MQTT, WebSockets, Edge Computing
Security: PKI, mTLS, OpenSSL, JWT, OAuth2, RBAC
DevOps & Cloud: Docker, Docker Compose, GitHub Actions (CI/CD), On-Premise Deployment, Grafana (Scenes), Linux, Azure, Vercel, Cloudflare R2 / AWS S3, Git
Databases & Testing: PostgreSQL, MongoDB, MySQL, Redis, Alembic, Pytest, Jest, Postman
Automation: Microsoft Power Automate, SharePoint, Power Apps

Education

Ramaiah University of Applied Sciences Bangalore, India
B.Tech in Computer Science and Engineering – CGPA: 9.4/10 *Dec 2021 – Jun 2025*